1 $A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$ $B^T = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$

$AB = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$ $BA = \begin{bmatrix} 7 & 6 \\ -2 & -1 \end{bmatrix}$

$(A+B)^T = \begin{bmatrix} 4 & 2 \\ 6 & 0 \end{bmatrix}$ $AB^T = \begin{bmatrix} 11 & -4 \\ 4 & -1 \end{bmatrix}$

trace $(A) = 4$
trace $(AB) = 6$

2 $M = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$

$\det(A - \lambda I) = 0$

$\det\left(\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right)$

$= \det\left(\begin{bmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{bmatrix}\right)$

$= (2-\lambda)(2-\lambda) + 1 = 0$

$4 - 4\lambda + \lambda^2 - 1 = 0$

$\lambda^2 - 4\lambda + 3$

$(\lambda - 1)(\lambda - 3) = 0$

eigenvalues $= 1$ and 3

for $\lambda = 1 \rightarrow \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \Rightarrow$ $x - y = 0$ w/ $x = 1$
$-x + y = 0$ $y = 1$

eigenvector $= (1, 1)$

for $\lambda = 3 \rightarrow \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \Rightarrow$ $-x - y = 0$ w/ $x = 1$
$y = -1$
eigenvector $= (1, -1)$

3 $\text{trace}(AB) = \text{trace}(BA)$

the diagonal of matrix $AB = A*B =$

$A(1,1)B(1,1) + A(1,2)B(2,1) \dots A(1,n)B(n,1)$

$\vdots$

$A(n,1)B(1,n) + A(n,2)B(2,n) \dots A(n,n)B(n,n)$

and diagonal of matrix $BA = B*A =$

$B(1,1)A(1,1) + B(2,1)A(1,2) \dots B(n,1)A(1,n)$

$\vdots$

$B(1,n)A(n,1) + B(2,n)A(n,2) \dots B(n,n)A(n,n)$

these are the same, with multiples reversed, the trace of each would be the sum along the diagonal

like: $A(1,1)B(1,1) + \dots + A(n,1)B(1,n) +$

$\vdots$

$+ A(1,n)B(n,1) + \dots + A(n,n)B(n,n)$

which is equal for both, $\therefore \text{trace}(AB) = \text{trace}(BA)$

4 verify invertible for $\begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$

$\det(C) = 24 - 14 = 10$ ✓

$\text{inverse} = \frac{1}{ad-bc}\begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} = \frac{1}{10}\begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$

$= \begin{bmatrix} 3/5 & -7/10 \\ -1/5 & 2/5 \end{bmatrix}$

Part 2

1 $P(X=k) = c \cdot \frac{3^k}{k!}$ w/ $k = 0, 1, 2 \ldots$

↑
discrete, use Σ

$\sum_{k=0}^{\infty} c \cdot \frac{3^k}{k!} = 1$

$1 = c \sum_{0}^{\infty} \frac{3^k}{k!}$ ← Taylor series for e^x

w/ $x = 3$ so $1 = c e^3$ $\boxed{c = \frac{1}{e^3}}$

b w/ $c = P(X=k) = \frac{1}{e^3}\left(\frac{3^k}{k!}\right)$

Poisson w/ $\lambda = 3$

$E[X] = 3 \quad Var(X) = 3$

2. $P(A \cap B) = P(A|B) \cdot P(B)$
$= 0.6 \cdot 0.3$
$= 0.18$

$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.18}{0.5} = 0.36$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$= 0.5 + 0.3 - 0.18$
$= 0.62$

3 w/ $\theta = P(H)$ and $1-\theta = P(T)$

For HHTHT we get:

$$\theta \cdot \theta \cdot (1-\theta) \cdot \theta \cdot (1-\theta)$$

$$= \theta^3 (1-\theta)^2$$

Take natural log:

$$= 3\ln\theta + 2\ln(1-\theta)$$

$$\frac{\partial L}{\partial \theta} = \frac{3}{\theta} - \frac{2}{1-\theta} = 0$$

$$3 - 3\theta = 2\theta$$

$$\theta = 3/5 = MLE$$

6 given $P(\theta) = \begin{cases} 2\theta & 0 \le \theta \le 1 \\ 0 & \text{otherwise} \end{cases}$

$$= \theta^3 (1-\theta^2) \cdot 2\theta$$

take natural log:

$$4\ln\theta + 2\ln(1-\theta)$$

$$\frac{\partial L}{\partial \theta} = \frac{4}{\theta} - \frac{2}{1-\theta} = 0$$

$$4 - 4\theta = 2\theta$$

$$\theta = \frac{2}{3}$$

4 $X = \{1, 3, 5, 7\}$ $\quad$ guess $= \frac{1}{\sqrt{2\pi\theta}} \exp\left(\frac{-(x-\mu)^2}{2\theta^2}\right)$

for n observations:

$$(2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x-\mu)^2\right)$$

take logs: $\quad -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x-\mu)^2$

$= \frac{5}{}$

$\sigma^2 = 4 \quad \sigma = 2 \quad \mu \sim N(0,1)$ Simulate $X \sim$

using $Exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ & $exp\left(-\frac{(5-\mu)^2}{8}\right)$

and prior $exp\left(-\frac{\mu^2}{2}\right)$

$\hookrightarrow exp\left(-\frac{(5-\mu)^2}{8} - \frac{\mu^2}{2}\right)$

$= \ln\left(exp\left(\frac{-(5-\mu)^2}{8} - \frac{\mu^2}{2}\right)\right) = \frac{-(5-\mu)^2}{8} - \frac{\mu^2}{2}$

$= \frac{2(5-\mu)}{8} - \frac{2\mu}{2} = 5 - \mu - 4\mu = 0$

$\boxed{\mu = 1}$

6 Prove $Var(x) = E(x^2) - E(x)^2$

$Var(x) = E[(x - E[x])^2]$

$= E[x^2 - 2xE[x] + E(x]^2]$

$= E[x^2] - 2E[x]E[x] + E[x]^2$

$= E[x^2] - E[x]^2$

Part 3:

$Var(u^T x) = E[(u^T x - E[u^T x])^2]$

but u^T is a constant in this case so

$E[u^T x] = u^T E[x]$

say $\mu = E[x]$

then

$Var(u^T x) = E[(u^T x - u^T \mu)^2]$

$= E[u^T (x - \mu))^2]$

$= E[u^T (x-\mu][u^T (x-\mu)^T])$

$Var(u^T x) = E[u^T (x-\mu)(x-\mu)^T u]$

$= u^T E[(x-\mu)(x-\mu)^T] u$

w $\Sigma = E[(x-\mu)(x-\mu)^T]$

$Var(u^T x) = u^T \Sigma u$